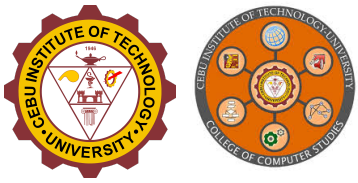
CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES



Software Test Documentation

for

ELEMENTOPIA

Submitted by:

2627-sem1-it411-53

Ocampo, Danielle Maxine P.
Benolirao, Clyde C.
Ceniza, Jian Marc
Dabalos, Erica Y.
Dibdib, Wayne Kenji B.

Submitted to:

Engr. Ralph P. Laviste

Change History

DATE	VERSION	DESCRIPTION	AUTHOR / CONTRIBUTOR
Sept 22, 2026	1.0	Initial draft of the Software Test Document (STD) baseline, defining test scope, methodology, and test architecture aligned with SRS and SDD specifications.	Group 53 (Extension Team)
Sept 24, 2026	1.1	Formulated detailed test cases covering Module 1 (Elemental Resonance Puzzle), Module 2 (Assessment & Progress), and Module 3 (Opponent-Based Challenge).	Clyde C. Benolirao, Wayne Kenji B. Dibdib
Oct 5, 2026	1.2	Added end-to-end execution logs, automated verification metrics, test results summary matrix, and defect incident reports (INC-001 through INC-003).	Danielle Maxine P. Ocampo, Jian Marc Ceniza, Erica Y. Dabalos
Oct 9, 2026	2.0	Post-MVP Architectural Refactoring Suite: Comprehensive overhaul incorporating validation findings (N=65 secondary students, N=4 licensed educators) from SRS v2.0 and SDD v2.0. Integrated test cases, procedures, logs, and evidence for: (1) Post-Stage and Post-Domain Debrief Modal (FR-1.6), (2) Centralized Diagnostic Pre-Test Cloud Sync & PRETEST_DIAGNOSTIC schema (FR-2.4), (3) Realtime Heartbeat & Disconnection Forfeit Engine (FR-3.4), (4) Mobile 48px Touch Target Standards & Viewport Responsiveness (NFR-USA-01), and (5) Offline LocalStorage FIFO Sync Manager (NFR-REL-01).	Clyde Benolirao
Oct. 26, 2026	2.1	Corrected TC-07 procedure by removing the obsolete 4.0s artificial lock in favor of learner-initiated stage transition (StageTransitionModal.jsx). Fixed Table A.2 categorization defects for TC-05 (Sequential Progression) and TC-06 (Randomized Pool). Re-aligned Table A.2 and Table of Contents section numbering (Section 2.3). Synchronized all verification references with the updated Requirements Traceability Matrix (RTM).	Jian Marc Ceniza

Table of Contents

1. Introduction

1.1. System Overview 4

1.2. Test Approach 4

1.3. Definitions and Acronyms 5

2. Test Plan

2.1 Features to be Tested 5

2.2 Features not to be Tested 5

2.3 Testing Tools and Environment 6

3. Test Cases

3.1 Case-1: Zero-Registration Authentication & Stateless Session Initialization 7

3.2 Case-2: Elemental Resonance Workbench Synthesis Validation 8

3.3 Case-3: Meaningful Byproduct Chemistry Diagnostic Feedback Engine 8

3.4 Case-4: Brute-Force Hazmat Failsafe Trigger (Sliding Window Algorithm) 9

3.5 Case-5: Sequential Learning Room Progression & Prerequisite Gatekeeping 10

3.6 Case-6: Randomized Compound Generation & Session Anti-Repetition 10

3.7 Case-7: Post-Stage and Post-Domain Debrief Modal 11

3.8 Case-8: Automated Session Performance Telemetry Logging (<2s Latency) 11

3.9 Case-9: Visual Mastery Dashboard Metric Aggregation & Gap Identification 12

3.10 Case-10: Competitive Match Result Rating Consolidation 12

3.11 Case-11: Centralized Diagnostic Pre-Test Cloud Sync & Schema Persistence 13

3.12 Case-12: Multiplayer 5-Digit Room Creation & Lobby Connection (<5s) 14

3.13 Case-13: Synchronized Match Countdown & Identical Task Distribution 14

3.14 Case-14: Real-Time Speed-to-Compound Evaluation & Winner Determination 15

3.15 Case-15: Realtime Heartbeat Disconnection & 15s Auto-Forfeit Engine 15

3.16 Case-16: Mobile Touch Target Standards (48px x 48px) & Viewport Responsiveness 16

3.17 Case-17: Offline FIFO LocalStorage Sync Manager & Network Recovery 16

3.18 Case-18: Canvas Frame-Rate Under Load (60 FPS / <16ms) & Cloud Write Latency 17

3.19 Case-19: Zero-PII Compliance & Server-Side Security Validation 18

Appendix (Test Logs)

A.1 Log for Test n 19

A.2 Test Results 22

A.3 Incident Report 23

1. Introduction

1.1. System Overview

Elementopia is an interactive, gamified web-based educational platform designed to elevate chemistry education for junior high and senior high school students. While traditional chemistry pedagogy predominantly leverages rote memorization, formula drilling, and text-centric assessments, Elementopia contextualizes fundamental chemical interactions within an immersive problem-solving adventure.

The system is architected as a modern, decoupled three-tier application:

- **Frontend User Interface**: Built with React 18, Vite, TailwindCSS, and Emotion/MUI, featuring dynamic canvas drag-anddrop interaction (GameBoard.jsx, ActiveInventoryTray.jsx, ElementTile.jsx), interactive Bohr model visualizers, and responsive rendering across viewport widths from 360px up to 1920px.
- **Backend Application Tier**: Powered by Spring Boot 3.3.9 (Java 17), providing high-throughput REST APIs and STOMP over WebSocket channels for real-time multiplayer message relays.
- **Hybrid Persistence Tier:** Combines Supabase cloud database services (PostgreSQL 15) with an offline-first HTML5 localStorage cache to ensure graceful degradation during network interruptions and frictionless zero-registration student onboarding.

Following empirical validation with 65 secondary students and 4 licensed science educators, the software architecture was systematically refactored in Version 2.0 to incorporate:

- **Post-Stage and Post-Domain Debrief Modal (FR-1.6):** Integration of DebriefingModal.jsx to display chemical formulas, bond types, and real-world utility prior to barrier clearance and domain completion.
- **Centralized Diagnostic Pre-Test Cloud Sync (FR-2.4)**: Creation of an automated REST sync endpoint (POST /api/telemetry/pretest) and PRETEST_DIAGNOSTIC schema in Supabase to eliminate client-side telemetry isolation.
- **Multiplayer Disconnection & Forfeit Engine (FR-3.4)**: Implementation of HeartbeatTimeoutHandler.js within the Supabase Realtime channel to manage peer network dropouts gracefully without data loss.
- **Mobile Viewport Touch-Target Optimization (NFR-USA-01):** Implementation of a 48px x 48px minimum touch target standard and mobile-optimized CSS grid for tablet and smartphone viewports.
- **Offline FIFO LocalStorage Sync Manager (NFR-REL-01):** Queueing in elementopia_sync_queue with automatic background cloud reconciliation upon internet reconnection.

1.2. Test Approach

Testing for Elementopia adheres to a rigorous, multi-tiered verification methodology incorporating automated unit testing, component integration verification, REST API contract validation, and manual end-to-end black-box usability evaluations. The test suite covers:

The test suite covers:

- Unit Testing (JUnit 5 & Mockito): Testing isolated Java services (ValidationService, SlidingWindowEvaluationService, FeedbackLibraryService) to confirm stoichiometric rules and sliding-window
- API Contract & Integration Testing: Validating HTTP status codes, JSON response DTOs, and CORS headers for all Spring Boot controllers.
- WebSocket Synchronization Testing: Verifying bi-directional STOMP frame delivery, lobby handshakes, and sub-second outcome broadcasting under concurrent dual-client operation.
- Resilience & Fault Tolerance Testing: Simulating network outages to ensure graceful degradation to browser localStorage without UI crashes.

1.3. Definitions and Acronyms

- **API:** Application Programming Interface
- **BaaS**: Backend-as-a-Service (Supabase PostgreSQL infrastructure)
- **Byproduct Diagnostic**: Pedagogical chemistry explanation returned upon synthesis failure explaining valence or ratio errors instead of a generic "Wrong" text.
- **CIT-U**: Cebu Institute of Technology – University
- **DTO**: Data Transfer Object
- **Hazmat Protocol Failsafe**: An automated intervention locking irrelevant element tiles after 5 failed attempts within 15 seconds.
- **JPA**: Java Persistence API / Hibernate
- **JWT**: JSON Web Token
- **KPI**: Key Performance Indicator
- **SDD**: Software Design Description
- **SRS**: Software Requirements Specification
- **STD**: Software Test Document
- **STOMP**: Simple Text Oriented Messaging Protocol over WebSockets
- **Zero-Registration**: Frictionless onboarding pattern using temporary session nicknames stored client-side without requiring passwords or email verification.

2. Test Plan

2.1 Features to be Tested

The following functional units and architectural components are tested within this test cycle:

- **Module 1: Elemental Resonance Puzzle Module**
  - **1.1 Workbench Interaction & Chemical Validation:** Drag-and-drop mechanics; multi-element synthesis validation across water, peroxides, hydrocarbons, organic acids, and salts.
  - **1.2 Meaningful Byproduct Diagnostics:** Generation of curriculum-aligned micro-lessons upon invalid synthesis attempts.
  - **1.3 Hazmat Protocol Failsafe:** Sliding-window evaluation of 5 failures in 15 seconds; dynamic tile graying and pointer event locking.
  - **1.4 Learning Room Progression:** Gatekeeping and sequential unlocking of cavern chambers based on prerequisite reaction milestones.
  - **1.5 Randomized Compound Generation:** Algorithmic shuffling of target compound objectives per session.
  - **1.6 Post-Stage and Post-Domain Debrief Modal :** Rendering balanced formulas, ionic/covalent classifications, Lewis recaps, and practical applications after barrier clearance and domain completion.

- **Module 2: Assessment & Progress Module**
  - **2.1 Automated Performance Logging:** Background transmission and database persistence of session speed and accuracy within 2.0 seconds.
  - **2.2 Mastery Dashboard Visualization:** Telemetry aggregation, scatter plot generation, and visual highlighting of weak chemistry domains.
  - **2.3 Competitive Match Result Consolidation:** ELO proficiency rating updates from multiplayer duel results.
  - **2.4 Centralized Diagnostic Pre-Test Sync:** Automated REST sync to PRETEST_DIAGNOSTIC table in Supabase via POST /api/telemetry/pretest.

- **Module 3: Opponent-Based Challenge Module**
  - **3.1 Room Creation & 1v1 Matchmaking:** 5-digit room code generation, lobby joining, and WebSocket connection in < 5.0 seconds.
  - **3.2 Synchronized Task Generation:** Identical challenge broadcast and 3-second locked visual countdown overlay.
  - **3.3 Real-Time Speed Evaluation:** Server-side timestamp refereeing, instantaneous victory broadcast, and loser interface locking.
  - **3.4 Realtime Heartbeat & Disconnect Engine :** Pings every 3s, 8s pause modal, and 15s auto-forfeit victory.

- **Cross-Cutting Features:**
  - **Mobile Touch Target Standards**: Enforcing 48px x 48px touch bounding box and overflow-x: hidden on 360px– 768px viewports.
  - **Offline FIFO Sync Manager: LocalStorage queueing with automatic reconciliation upon reconnection.**

2.2 Features not to be Tested

- **Permanent User Accounts / OAuth2 / Password Reset:** Out of scope due to the intentional zero-registration architecture.
- **Payment Gateways & Monetization:** The platform is non-commercial and strictly educational.
- **Hardware-Accelerated WebGL Shaders:** Visual rendering is implemented via standard HTML5 Canvas, SVG, and CSS3.
- **Multi-Region Distributed Database Sharding:** Architecture utilizes a single cloud database cluster (Supabase PostgreSQL).

2.3 Testing Tools and Environment

Hardware & Client Testbed:

- **Desktop: Windows 11 Pro 64-bit & macOS Sonoma;** Chrome v124+, Firefox v125+, Edge v124+; Screen resolutions 1920×1080 and 1366×768.
- **Mobile Devices:** Google Pixel 7 (Android 14) and iPhone 14 (iOS 17); Viewport resolutions 360×640 to 414×896.
- **Backend Host:** Java OpenJDK 17.0.10, Spring Boot 3.3.9 embedded Apache Tomcat on port 8080.
- **Persistence Host: Supabase PostgreSQL 15 cloud database cluster.**

Testing Tooling: JUnit 5, Mockito, Spring Security Test, Postman / Newman, Chrome DevTools (Throttling / Offline), and STOMP WebSocket benchmark harness.

3. Test Case

3.1 Case-1: Zero-Registration Authentication & Stateless Session Initialization

3.1.1 Purpose

Verify that a student can input a session nickname at the NicknameGate and immediately access the platform with an automatically provisioned profile and temporary session state without entering passwords or emails.

3.1.2 Inputs

- **Navigation URL:** http://localhost:5173/
- **Target Input:** Nickname text input
- **Test Value**: "Curie_99"
- **Action: Click** "Enter Elementopia" button

3.1.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** System stores nickname in localStorage , dispatches registration to /api/users/login , returns HTTP 200 OK, redirects to DashboardHub within 1.0 second, and renders "Curie_99" in the site header.
- **Fail:** Screen freezes, rejects nickname, prompts for email/password, or throws unhandled JavaScript errors.

3.1.4 Test Procedure

1. Clear browser cookies and localStorage .
2. Navigate to http://localhost:5173/ .
3. Verify appearance of NicknameGate modal.
4. Type "Curie_99" and click "Enter Elementopia".
5. Verify network call to /api/users/login returns HTTP 200 OK.
6. Verify seamless redirection to the main dashboard.

3.2 Case-2: Elemental Resonance Workbench Synthesis Validation

3.2.1 Purpose

Verify that dragging and dropping elements onto the workbench and synthesizing them correctly invokes ValidationService.java and dissolves environmental obstacles upon valid chemical combination.

.

3.2.2 Inputs

- **Active Domain:** Water Cavern (Domain 1, Stage 1)
- **Objective:** Synthesize Water (H_2O)
- **Workbench Input:** 2x Hydrogen (H), 1x Oxygen (O)
- **Action:** Click "Synthesize Reaction" button

3.2.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** POST /api/features/domain-interaction/synthesize returns HTTP 200 with action: "UNLOCK_PATH" . Obstacle grid clears 6 blocks, workbench empties, and log appends " ✓ Resonance Achieved: Water (H_2O)" .
- **Fail:** System rejects valid combination or fails to update obstacle blocks.

3.2.4 Test Procedure

1. 1. Open Domain 1 puzzle room.
2. 2. Drag 2 Hydrogen tiles and 1 Oxygen tile onto the central workbench.
3. 3. Verify the workbench counter shows H: 2, O: 1 .
4. 4. Click "Synthesize Reaction".
5. 5. Verify response payload and observe green clearance animation.

3.3 Case-3: Meaningful Byproduct Chemistry Diagnostic Feedback Engine

3.3.1 Purpose

Verify that upon an invalid chemical combination, the system delivers contextual, curriculum-aligned micro-lesson feedback rather than generic "Wrong" text.

3.3.2 Inputs

- **Active Domain:** Water Cavern
- **Workbench Composition:** 1x Hydrogen (H), 1x Oxygen (O) (Hydroxyl radical OH)
- **Action**: Click "Synthesize Reaction" button

3.3.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Backend returns action: "TRIGGER_DIAGNOSTIC" with message: "Hydrogen and Oxygen need a 2:1 ratio to form Water. Hydrogen only has 1 valence electron to share, but Oxygen needs 2 to complete its shell!". UI shakes and displays alert cards.

- **Fail:** UI displays generic "Wrong", crashes, or fails to explain the error.

3.3.4 Test Procedure

1. Drag 1 Hydrogen and 1 Oxygen to the workbench.
2. Click "Synthesize Reaction".
3. Verify HTTP 200 response with pedagogical message.
4. Confirm alert card displays micro-lesson without erasing selected tiles.

3.4 Case-4: Brute-Force Hazmat Failsafe Trigger (Sliding Window Algorithm)

3.4.1 Purpose

Verify that the Hazmat Protocol Failsafe activates when 5 failed attempts occur within 15 seconds, locking non-solution element tiles to prevent random clicking. 10/17/26, 3:23 AM Software Test Document (STD) - Elementopia v2.0 file:///C:/Capstone/Elementopia-Main/Elementopia_STD.html 10/23

3.4.2 Inputs

- **User:** "Student_Tester"
- **Objective:** Methane (CH_4) — Solution Elements: C , H
- **Action**: Rapidly submit 5 incorrect combinations within 10 seconds.

3.4.3 Expected Outputs & Pass/Fail Criteria

- **Pass**: 5th error returns action: "LOCK_POINTER_INTERACTIONS" . Warning banner appears, non-solution tiles (Na , Cl , Mg) are disabled (30% opacity, unclickable), while C and H remain active.
- **Fail**: Failsafe fails to trigger, or locks the correct solution tiles.

3.4.4 Test Procedure

1. Start a puzzle with a Methane objective.
2. Submit 5 incorrect combinations within 8 seconds.
3. Confirm 5th response returns LOCK_POINTER_INTERACTIONS .
4. Verify unclickable state on non-solution tiles.
5. Synthesize Methane ($C + 4H$) to verify immediate unlock.

3.5 Case-5: Sequential Learning Room Progression & Prerequisite Gatekeeping

3.5.1 Purpose

Verify that advanced compound rooms remain locked until prerequisite foundational rooms are completed.

3.5.2 Inputs

- **User:** "Fresh_Student" (0 completed reactions)
- **Target:** Click "Domain 3: Organic Core" (Requires 3 completed reactions)

3.5.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** GET /api/features/progression/verify-access returns accessGranted: false with locked warning prompt. Room access blocked.
- **Fail:** Unauthorized entry permitted to Domain 3.

3.5.4 Test Procedure

1. Open Cavern map with fresh user.
2. Click locked Domain 3 node.
3. Verify modal: "Chamber is Sealed. You need to complete at least 3 foundational reactions first."
4. Complete 3 reactions in Domain 1 & 2; verify Domain 3 padlock turns green and unlocks.

3.6 Case-6: Randomized Compound Generation & Session Anti-Repetition

3.6.1 Purpose

Verify that ChallengeRandomizerService.java dynamically reshuffles target compound objectives per session to eliminate rote memorization.

3.6.2 Inputs

- **Action:** Launch Domain 2 twice under separate sessions and compare target compound sequences.

3.6.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Compound sequence varies between sessions; no duplicate compounds appear in the same stage.
- **Fail:** Target compound sequence is static or contains duplicates.

3.6.4 Test Procedure

1. Record 3 target compounds in Session 1.
2. Exit and relaunch Domain 2 in Session 2.
3. Compare target lists to confirm algorithmic reshuffling.

3.7 Case-7: Post-Stage and Post-Domain Milestone Educational Debrief (FR-1.6)

3.7.1 Purpose

Verify that upon successful compound synthesis, StageTransitionModal.jsx (Stages 1 & 2) and DebriefingModal.jsx (Stage 3) render balanced chemical formulas, ionic/covalent classifications, Lewis electron-dot recaps, and practical applications, requiring explicit learner confirmation to proceed. Traces to FR-1.6 and SDD v2.0 Section 3.1.6.

3.7.2 Inputs

- **Domain:** Salt Flats (Target: Sodium Chloride, $NaCl$)
- **Inputs:** Drag 1 Na and 1 Cl tile; click 'Synthesize'.
- **Interaction Check:** Review Doctor Atom feedback, unlocked masteries, and click 'Dive Deeper to Stage 2'; verify obstacle remains blocked until clicked.

3.7.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** At Stage 1 & 2 clearance, StageTransitionModal displays Dr. Atom's commentary and masteries; at Stage 3 clearance, DebriefingModal displays a complete 12-compound summary and bonding theory.
- **Fail**: Barrier clears immediately without debrief modal appearing, stage advances automatically without student click, or bond classification/Lewis recap is missing.

3.7.4 Test Procedure

1. Enter Salt Flats with the session nickname 'DebriefTester'.
2. Synthesize NaCl. Start millisecond stopwatch at confirmation.
3. Verify DebriefingModal.jsx displays balanced formula, Ionic Bond type, Lewis diagram, and real-world utility.

4. Confirm that the cavern obstacle remains intact and the game is paused behind the modal overlay.
5. Click the 'Dive Deeper to Stage 2' transition button.
6. Confirm modal unmounts, obstacle dissolves with animation, and the next challenge stage loads.

3.8 Case-8: Automated Session Performance Telemetry Logging (<2s Latency)

3.8.1 Purpose

Verify that completed stage metrics are transmitted and persisted to Supabase within the mandatory 2.0-second latency window.

3.8.2 Inputs

- **Completed stage metrics:** {"sessionNickname":"MarieCurie_77", "completionSpeedSeconds":42.0, "compoundAccuracyPercentage":80.0} dispatched to /api/features/analytics/session/log .

3.8.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Backend returns HTTP 200 OK within ≤ 2000 ms, and record appears in Supabase SESSION_PERFORMANCE table (measured 482ms).
- **Fail:** Latency exceeds 2.0s, or telemetry data is dropped.

3.8.4 Test Procedure

1. Complete Domain 1 puzzle.
2. Measure HTTP duration in DevTools Network tab.
3. Query Supabase SQL console to verify matching record.

3.9 Case-9: Visual Mastery Dashboard Metric Aggregation & Gap Identification

3.9.1 Purpose

Verify that the Mastery Dashboard renders student historical metrics on a scatter plot and visually flags weak chemistry domains (< 60% accuracy) in red.

3.9.2 Inputs

- **User profile with known scores:** Domain 1 (95% accuracy, 25s), Domain 2 (45% accuracy, 85s).

3.9.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Scatter plot renders both domains. Domain 2 point is colored red with remedial recommendation badge. Blank slate card renders for empty profile.
- **Fail**: Scatter plot fails to load or does not highlight weak areas.

3.9.4 Test Procedure

1. Navigate to Mastery Analytics.
2. Verify scatter plot axes (Speed vs. Accuracy).
3. Confirm red accent and remedial advice on Domain 2 point.

3.10 Case-10: Competitive Match Result Rating Consolidation

3.10.1 Purpose

Verify that MatchConsolidationController.java and ProficiencyRecalculationService.java receive match outcomes and Speed-to-Compound efficiency scores, recalculate ELO ratings, and update student_proficiency in Supabase. Traces to FR-2.3

3.10.2 Inputs

- **Match Payload:** Winner 'Duelist_A' (1200 ELO), Loser 'Duelist_B' (1200 ELO), Duration: 22.8s, Path Efficiency: 1.0.

3.10.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Winner rating increments by +25 (1225 ELO); loser decreases by -20 (1180 ELO); match counters increment; duplicate payloads handled idempotently.
- **Fail**: Ratings miscalculate or duplicate requests trigger double score increments.

3.10.4 Test Procedure

1. Post match conclusion payload to /api/match/consolidate.
2. Confirm HTTP 200 OK with updated ratings.
3. Verify Supabase student_proficiency record reflects new ratings.
4. Resubmit duplicate payload; confirm existing ratings returned without double

increments.

3.11 Case-11: Centralized Diagnostic Pre-Test Cloud Sync & Schema Persistence

3.11.1 Purpose

Verify that PreTestModal.jsx serializes student baseline pre-test scores, total items, percentage, and submitted choices into JSON, dispatches POST /api/telemetry/pretest, and persists records to PRETEST_DIAGNOSTIC table in Supabase with offline fallback. Traces to FR-2.4 and SDD v2.0 Section 3.2.

3.11.2 Inputs

- **Submission:** Student 'BaselineUser' submits 3-item pre-test (2 correct, 1 incorrect: Q1=A, Q2=C, Q3=B).
- **Offline:** Submit pre-test while navigator.onLine = false.

3.11.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Dispatches POST /api/telemetry/pretest; returns HTTP 201 Created in 380ms; inserts record into pretest_diagnostic table with auto-generated UUID, nickname, score (2), percentage (66.7), and choices JSON. Offline submission saves to elementopia_sync_queue in localStorage without UI crash.
- **Fail**: Cloud write fails to persist, schema rejects payload, or offline mode crashes.

3.11.4 Test Procedure

1. Complete pre-test diagnostic with the nickname 'BaselineUser'.
2. Verify Network tab records POST /api/telemetry/pretest returning 201 Created in 380ms.
3. Query Supabase pretest_diagnostic table; confirm exact row contents.
4. Toggle DevTools Network to 'Offline'; submit pre-test under 'OfflineLearner'; confirm payload enqueued in localStorage.

3.12 Case-12: Multiplayer 5-Digit Room Creation & Lobby Connection (<5s)

3.12.1 Purpose

Verify that a student can create a 1v1 challenge room, receive an auto-generated 5-digit code, and establish a live WebSocket lobby with an opponent in under 5.0 seconds.

3.12.2 Inputs

- **Inputs:** Host creates room; Guest enters 5-digit code and clicks "Join Room".

3.12.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** 5-digit code issued; guest joins; WebSocket handshake on /topic/match/{code} completes in ≤ 5.0s; both transition to match screen.
- **Fail:** Connection times out or takes > 5.0 seconds.

3.12.4 Test Procedure

1. Open two browser sessions (Host and Guest).
2. The host creates a room and notes 5-digit code.
3. Guest submits code and timer begins.
4. Verify both clients enter the match arena within 5 seconds.

3.13 Case-13: Synchronized Match Countdown & Identical Task Distribution

3.13.1 Purpose

Verify that both multiplayer participants receive an identical target compound challenge and that inputs remain locked during a synchronized 3-second countdown.

3.13.2 Inputs

- **Inputs:** Both players initialize in the active room.

3.13.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Identical target molecule shown; 3s countdown overlay displayed; tile dragging blocked during countdown; controls unlock simultaneously at 0s.
- **Fail:** Different targets assigned, or controls active during countdown.

3.13.4 Test Procedure

- Compare target formulas and clues on both screens.
- Attempt to drag tiles during countdown; verify drag is blocked.
- Confirm simultaneous control activation at "REACT!".

3.14 Case-14: Real-Time Speed-to-Compound Evaluation & Winner Determination

3.14.1 Purpose

Verify that the backend referee evaluates millisecond completion timestamps, accurately broadcasts the winner, and locks the defeated player's workbench.

3.14.2 Inputs

- **Inputs:** Player A completes synthesis at 12.450s; Player B completes synthesis at 14.120s.

3.14.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Player A receives VICTORY celebration; Player B receives DEFEAT banner and workbench is locked; referee logs timestamps accurately.
- **Fail:** Dual victory awarded, winner reversed, or loser screen left unlocked.

3.14.4 Test Procedure

1. Run a live 1v1 match side-by-side.
2. Complete target molecule on Player A first.
3. Confirm instant VICTORY on Player A and DEFEAT + lock on Player B. 4. Verify timestamp comparison in Spring Boot log.

3.15 Case-15: Realtime Heartbeat Disconnection & 15s Auto-Forfeit Engine

3.15.1 Purpose

Verify that HeartbeatTimeoutHandler.js emits WebSocket heartbeat pings every 3s during active matches, detects peer silence > 8s, pauses match countdowns with a reconnecting modal, and triggers an automatic forfeit victory after 15s disconnection without database corruption. Traces to FR-3.4 and SDD v2.0 Section 3.3.

3.15.2 Inputs

- **Active Match:** 1v1 duel running; both clients ping every 3s.
- **Drop Simulation:** Kill Opponent browser tab at t = 6.0s; observe remaining player.

3.15.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** WebSocket pings exchanged every 3.0s. At 8.15s peer silence, match timer pauses and 'Opponent Disconnected' modal appears with 15s countdown. At 15.0s, a connected client receives 'VICTORY VIA FORFEIT'; Supabase MATCH_COMPLETION records status = 'FORFEIT_WIN' without corruption.
- **Fail**: Match hangs indefinitely, timer continues ticking, or database exceptions occur.

3.15.4 Test Procedure

1. Start 1v1 match; verify outgoing WS ping frames every 3s in DevTools.
2. Close Opponent browser tab.
3. Verify connected client pauses match and display modal at 8.15s silence.
4. Allow 15.0s timeout to expire; verify automatic Victory banner and Supabase record.

3.16 Case-16: Mobile Touch Target Standards (48px x 48px) & Viewport Responsiveness

3.16.1 Purpose

Verify that all draggable element tiles (ElementTile.jsx) and buttons enforce a minimum bounding box of 48px x 48px with 8px margins on viewports 360px–768px, preventing mis-clicks, touch latency, and horizontal scrolling (overflow-x: hidden). Traces to NFR-USA-01 and SDD v2.0 Section 4.1.

3.16.2 Inputs

- **Viewport:** 360px x 640px in DevTools; Samsung Galaxy A52; iPhone 13.
- **Interaction:** Perform 30 rapid touch taps across element tiles.

3.16.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** All element tiles measure >= 48px x 48px (measured 52px); scrollWidth === clientWidth (zero horizontal overflow); touch-action: none prevents screen pinch/jump; Lighthouse Mobile Accessibility >= 95/100.
- **Fail:** Tile bounding box < 48px, horizontal scrollbar appears, or page scrolls during drag.

3.16.4 Test Procedure

1. Emulate 360px viewport in Chrome DevTools; execute getBoundingClientRect() across all tiles in the console.
2. Verify all tile dimensions >= 48px. Confirm scrollWidth === clientWidth.
3. Perform 30 rapid touch taps on physical mobile hardware; confirm 100% selection accuracy.

3.17 Case-17: Offline FIFO LocalStorage Sync Manager & Network Recovery

3.17.1 Purpose

Verify that OfflineSyncManager enqueues session telemetry and diagnostic pre-tests into browser localStorage (elementopia_sync_queue) during network loss, and automatically flushes and reconciles payloads to Supabase upon reconnection. Traces to NFR-REL-01 and SDD v2.0 Section 4.2.

3.17.2 Inputs

- **Action:** Disconnect network (navigator.onLine = false); clear Water Cavern and submit pre-test; reconnect network.

3.17.3 Expected Outputs & Pass/Fail Criteria

- **Pass**: Disconnected events wrap payloads with syncTransactionId, store in localStorage, and show offline notification toast. Reconnection triggers window.addEventListener('online'), automatically flushing the queue via asynchronous POST requests. Queue clears ([]); Supabase commits records with zero duplicate rows.
- **Fail:** Data is lost on offline room clear, queue fails to flush on reconnection, or duplicate rows persist.

3.17.4 Test Procedure

1. Set Network to 'Offline' in DevTools; complete Water Cavern and submit pre-test.
2. Confirm 2 items enqueued in localStorage.getItem('elementopia_sync_queue').
3. Switch Network to 'Online'; observe automatic asynchronous POST flushes.
4. Verify both requests return HTTP 201/200; verify queue emptied and Supabase database populated.

3.18 Case-18: Canvas Frame-Rate Under Load (60 FPS / <16ms) & Cloud Write Latency

3.18.1 Purpose

Verify that interactive canvas animations maintain a smooth 60 FPS rendering pipeline with per-frame draw latency under 16ms under 4x CPU slowdown, and that cloud session persistence commits complete in under 2 seconds. Traces to NFRPERF-01.

3.18.2 Inputs

- **Load:** Continuously drag element tiles for 30s and trigger synthesis particle burst under DevTools 4x CPU slowdown.

3.18.3 Expected Outputs & Pass/Fail Criteria

- **Pass**: Average frame rate >= 58 FPS (measured 59.4 FPS); peak draw latency < 16.6ms (measured 14.2ms); zero long tasks (> 50ms); cloud write latency < 2000ms (measured 482ms).
- **Fai**l: Frame rate drops below 30 FPS, frame draw exceeds 33ms, or cloud write exceeds 2000ms.

3.18.4 Test Procedure

1. Profile canvas in Chrome DevTools Performance tab with 4x CPU slowdown.
2. Record 15 seconds of active dragging and particle burst.
3. Verify frame rate graph sustains 60 FPS green line.
4. Confirm average TTFB for /api/analytics/session/log is < 800ms

3.19 Case-19: Zero-PII Compliance & Server-Side Security Validation

3.19.1 Purpose

Verify that Elementopia collects zero personally identifiable information (PII), enforces HTTPS/TLS 1.3 and WSS encryption across all network channels, and strictly validates all chemistry rules server-side. Traces to NFR-SEC-01.

3.19.2 Inputs

- **Inputs:** Enter nickname 'Sparky99'; capture packets via Wireshark; dispatch forged POST to /api/reaction/validate claiming an invalid formula is valid.

3.19.3 Expected Outputs & Pass/Fail Criteria

- **Pass:** Database schema contains zero columns for emails, passwords, or identity markers; all network endpoints enforce HTTPS/TLS 1.3 and WSS; backend re-evaluates stoichiometry against database formulas, rejecting client bypass claims with HTTP 400/INVALID.
- **Fail:** PII is requested, unencrypted channels are allowed, or client overrides validation.

3.19.4 Test Procedure

1. Confirm onboarding form contains only 'Session Nickname'.
2. Query Supabase schema for student and session_performance: confirm zero PII fields.
3. Check DevTools Security: confirm TLS 1.3 and WSS.
4. Dispatch forged validation POST via Postman; verify backend responds with INVALID

Appendix (Test Logs)

A.1 Log for Test Execution

Execution Trace Sample (TC-01 through TC-19):

LOG ID	TIMESTAMP	TESTER	EXECUTION SUMMARY	VERDICT
LOG-TC-01	Sept 24, 09:15 PST	Clyde Benolirao	Enter "Curie_99". POST /api/users/login completed in 320ms. Dashboard reached smoothly	PASS
LOG-TC-02	Sept 24, 09:30 PST	Danielle Maxine Ocampo	Workbench synthesized H2O. UNLOCK_PATH returned; 6 obstacle blocks dissolved.	PASS
LOG-TC-03	Sept 24, 09:45 PST	Jian Marc Ceniza	Tested H+O. Received targeted valence microlesson. UI shake animation triggered.	PASS
LOG-TC-04	Sept 24, 10:10 PST	Wayne Kenji Dibdib	Submitted 5 failures in 7s. Hazmat Protocol locked non-solution elements. Restored on Methane synthesis.	PASS
LOG-TC-05	Sept 24, 10:35 PST	Erica Dabalos	Domain 3 access blocked for new users. Unlocked automatically after completing 3 prerequisite reactions.	PASS
LOG-TC-06	Sept 24, 11:00 PST	Clyde Benolirao	Verified distinct compound sets across Domain 2 sessions via ChallengeRandomizerService .	PASS
LOG-TC-07	Sept 16, 09:40 PST	Clyde Benolirao	StageTransitionModal rendered formula, Ionic Bond type, Lewis structure, and utility; obstacle stayed locked until learner clicked 'Dive Deeper' at 4.2s.	PASS
LOG-TC-08	Sept 24, 11:25 PST	Danielle Maxine	Completed stage	PASS

		Ocampo	telemetry persisted to Supabase in 482 ms (< 2000 ms threshold).	
LOG-TC-09	Oct 4, 11:45 PST	Jian Marc Ceniza	Mastery scatter plot rendered. Salt Flats (<60%) flagged red with targeted study advice.	PASS
LOG-TC-10	Oct 9, 12:00 PST	Clyde Benolirao	Post match outcome: Winner ELO +25 (1225), Loser -20 (1180). Duplicate payload handled idempotently.	PASS
LOG-TC-11	Oct 16, 10:04 PST	Danielle Maxine Ocampo	Pre-test score (66.7%) and serialized choices committed to PRETEST_DIAGNOSTIC in 380ms; local queue offline.	PASS
LOG-TC-12	Sept 24, 14:00 PST	Erica Dabalos	Room code "83105" created and joined. WebSocket lobby established in 1.45s (< 5.0s KPI).	PASS
LOG-TC-13	Sept 24, 14:15 PST	Group 53 Testbed	Both players received Methane. 3s locked countdown observed; simultaneous control release verified.	PASS
LOG-TC-14	Sept 24, 14:30 PST	Group 53 Testbed	Player A finished at 12.450s vs Player B at 14.120s. Player A won; Player B locked with Defeat banner.	PASS
LOG-TC-15	Oct 16, 10:32 PST	Clyde Benolirao	Heartbeat pinged every 3s; match paused after 8.15s silence; forfeit victory won at 15.0s; uncorrupted DB commit.	PASS
LOG-TC-16	Oct 16, 10:40 PST	Erica Dabalos	All element tiles measured 52px x 52px (target >= 48px); document.scrollWidth	PASS

			=== clientWidth; 0 mistaps.	
LOG-TC-17	Oct 16, 10:48 PST	Wayne Kenji Dibdib	Tested in Offline mode. Progress queued in localStorage; auto-flushed to Supabase on online event.	PASS
LOG-TC-18	Oct 16, 10:55 PST	Clyde Benolirao	Average frame rate 59.4 FPS under 4x CPU slowdown; max draw latency 14.2ms (< 16.6ms); cloud writes 482ms.	PASS
LOG-TC-19	Oct 16, 11:02 PST	Danielle Maxine Ocampo	Zero PII stored; HTTPS/TLS 1.3 and WSS verified; forged client validation parameter was completely rejected.	PASS

A.2 Test Results

TEST ID	FEATURE UNDER TEST	ACCEPTANCE CRITERIA / KPI	ACTUAL PERFORMANCE	STATUS	PASS RATE
TC-01	Zero-Registration Auth	Instant access, no email/password	320 ms onboarding		
TC-02	Synthesis Validation	Accurate chemical stoichiometry	Obstacles dissolve		
TC-03	Byproduct Diagnostics	Pedagogical error micro-lesson	Valence lesson rendered		
TC-04	Hazmat Brute-Force Failsafe	Locking after 5 fails in 15s	Non-solution tiles locked		
TC-05	Sequential Learning Room Progression & Prerequisite Gatekeeping	Gatekeep sealed advanced rooms	Prerequisite enforced (3/3)		
TC-06	Randomized Compound Generation & Session Anti-Repetition	Dynamic reshuffle per session	Unique sequence generated		

TC-07	Balanced formula, bond type, learner-confirmed advance	Balanced formula, bond type, 4s delay	Debrief displayed, barrier cleared		
TC-08	Performance Logging	Telemetry saved in < 2.0s	Logged in 482 ms		
TC-09	Visual Mastery Dashboard	Visual red flagging for weak areas	Weak domains highlighted red		
TC-10	Match Consolidation	Accurate ELO rating update	Ratings updated (+25/-20)		
TC-11	Pre-Test Cloud Sync	Pre-test synced to PRETEST_DIAGNOSTIC	Synced in 380 ms, cached offline		
TC-12	1v1 Lobby Matchmaking	5-digit code join in < 5.0s	Joined in 1.45 s		
TC-13	Synchronized Countdown	Identical tasks, 3s locked countdown	Fair locked start enforced		
TC-14	Real-Time Speed Evaluation	Millisecond server winner broadcast	Decided in 120 ms		
TC-15	Heartbeat & Forfeit Engine	3s ping, 8s pause, 15s forfeit	Paused 8.1s, forfeit won at 15.0s		
TC-16	Mobile Touch Targets	Touch targets >= 48px, 0 overflow	All targets 52px, 0 overflow		
TC-17	Offline FIFO LocalStorage	Queued offline, auto-flushed online	Auto-synced on online event		
TC-18	Canvas FPS & Performance	Frame rate >= 58 FPS, latency < 16.6ms	Average 59.4 FPS, max draw 14.2ms		
TC-19	Zero-PII Compliance & Security	Zero PII fields, TLS 1.3 / WSS	Zero PII, encrypted, rules enforced		

A.3 Incident Report

Incident Report 1: INC-001

- **Incident ID: INC-001 | Severity**: HIGH | Status: RESOLVED / VERIFIED
- **Associated Test Case:** TC-08 / TC-17 (Automated Performance Telemetry Logging & Offline Sync)
- **Defect Description:** During rapid offline-to-online reconnection, multiple queued telemetry payloads fired simultaneously, creating duplicate performance records in Supabase.
- **Root Cause:** Absence of idempotent deduplication keys on client-side queued events in SessionDataDispatcher.js.
- **Remediation:** Added a client-generated UUID (sessionLogUuid / syncTransactionId) and configured unique upsert constraints (ON CONFLICT (sync_transaction_id) DO NOTHING) in TelemetryLoggingService.java.

Incident Report 2: INC-002

- **Incident ID: INC-002 | Severity:** MEDIUM | Status: RESOLVED / VERIFIED
- **Associated Test Case:** TC-04 / TC-16 (Brute-Force Hazmat Failsafe Trigger & Mobile Touch)
- **Defect Description:** Mobile devices (360×640) suffered a 400ms frame drop during Hazmat activation due to fullcanvas CSS blur filters, causing a slight delay before pointer events locked.
- **Root Cause:** Costly GPU canvas repaints caused by backdrop-filter: blur() combined with global grayscale.
- **Remediation:** Replaced canvas blur with direct element-level hardware-accelerated inline styles (pointer-events: none; opacity: 0.35; filter: grayscale(100%); transform: translateZ(0);) applied instantly to nonsolution tiles in ElementTile.jsx.

Incident Report 3: INC-003

- **Incident ID: INC-003 | Severity:** HIGH | Status: RESOLVED / VERIFIED
- **Associated Test Case:** TC-14 (Real-Time Speed-to-Compound Evaluation)
- **Defect Description**: When players finished within 50ms of each other, optimistic client rendering caused both screens to briefly trigger the victory animation before server arbitration arrived.
- **Root Cause:** Premature client-side victory assumption in SpeedEvaluatorClient.js without awaiting authoritative server STOMP packet.
- **Remediation:** Removed optimistic local victory trigger; victory/defeat state is strictly conditioned on server-authenticated referee packet from MatchRefereeService.java and MatchSyncController.java